

PATH 1017

Anemia

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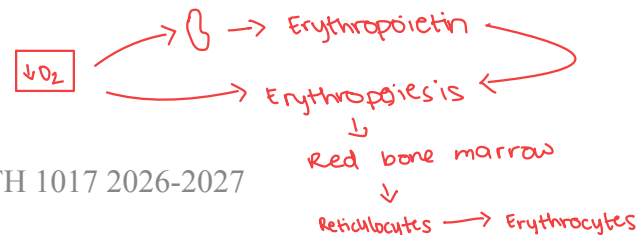
Anemia

- Abnormally low level of:
 - Hemoglobin
 - Circulating RBCs
 - Or both
- Leads to:
 - Decreased oxygen carrying capacity of blood

BLOOD

- ↳ can't replicate
- ↳ can't regenerate

↳ Bone marrow release blood when O_2 is low



Anemia can result from...

Impaired
production of
RBCs

- Nutritional deficiencies
- Aplastic

Increased
destruction of
RBCs

- Hemolytic
 - Sickle cell
 - Thalassemia

Massive or
chronic blood
loss

- Trauma
- GI bleed



Test → Don't memorize manifestations
understand why it happens

General manifestations are caused by:

1. ↓hemoglobin
2. Hypoxia from deficient O₂ transport
3. Compensatory mechanisms attempting to increase O₂ delivery

• List the common manifestations related to each:

- fatigue 2
- tachycardia 3
- paleness 3
- cyanosis 2
- chest pain/Angina 2
- Dizziness
- cold

Diagnostic Tests

- **Complete Blood Count (CBC)**

- Hemoglobin [Hgb]
- Hematocrit (Hct)
- RBC count
- RBC size (MCV) & RBC colour (MCHC) indices are used to differentiate anemia types

- **Blood Smear**

- Stained smear of blood viewed under microscope

- **Bone Marrow Aspiration**

- If blood smear abnormal
- Sites: posterior Iliac crest or sternum

CBC → MCV and MCHC

MCV (mean corpuscular volume)

- ↓ MCV = microcytic (small cell) anemia
- ↑ MCV = macrocytic (large cell) anemia
- Normal MCV = Normocytic anemia

MCHC (mean corpuscular hemoglobin concentration)

- ↓ MCHC/colour = hypochromic
- ↑ MCHC/colour = hyperchromic
- Normal MCHC/colour = normochromic

Additional diagnostic tests for specific anemias

- Iron level:
 - Ferritin,
 - Total iron-binding capacity (TIBC)
- Serum B12 level
- Gastric acid levels
- Antibodies to intrinsic factor
- Schilling test (B12 excretion)

Which type of anemia do these test help to diagnose?

Anemias of Deficient RBC Production

Iron deficiency anemia

- Often caused by blood loss
- Dietary deficiency

Megaloblastic anemias

Vitamin B12-deficiency (pernicious anemia)
Impaired DNA synthesis = enlarged red cells

Aplastic anemia

- Bone marrow depression

Chronic disease anemias

- Chronic inflammation
- Chronic renal failure
- Chronic infections

Iron-Deficiency Anemia

- Decreased iron present required for hemoglobin synthesis.
 - Decreased oxygen carrying capacity
- RBCs are:
 - Hypochromic & microcytic
 - Poikilocytosis (irregular shape)
 - Anisocytosis (irregular size)

What lab results are expected?

↓ Hgb

↓ MCV - MCH

Iron-Deficiency Anemia

Etiology:

- Chronic blood loss,
- Deficient diet
- Increased demands

Treatment:

- Improved diet
- Iron supplements
- Identification of cause of blood loss

What are some possible causes in:

- Women?

* long mens cycle

- Adolescents?

- Children?

- Medication-related?

Iron-Deficiency Anemia

Symptoms

- **Fatigue**
- Lethargy
- Weakness
- **Pallor**
- Cold intolerance
- Loss of concentration
- Irritability
- Loss of appetite

- Chronic State
 - **Tachycardia**
 - **SOB**
 - Hypotension
 - Brittle nails
 - Spoon-shaped fingernails
 - Stomatitis (mouth inflammation)
 - Glossitis (tongue inflammation)
 - Fissures in corners of mouth
 - Pica

Vitamin B12 Deficiency Anemia

- **Megaloblastic anemia**
- Erythrocytes are large(macrocytic), often with oval shape
- Poikilocytosis and teardrop shapes

Why is vitamin B12 necessary for RBCs?

↳ cells can't survive

How does lack of B12 affect RBCs?

What lab results are expected?

↑ MCV
Normal MCH

Is pernicious anemia
the same as Vit B12
deficiency anemia

B12-deficiency anemia

↳ Diet related

- Etiologies:
 - Vitamin B12 deficiency
 - Pernicious anemia (from atrophic gastritis)
 - lack of intrinsic factor in the stomach (antibodies often present)
- Treatment:
 - Vitamin B12 injections regularly
- Prognosis: Good with treatment

What is the difference between these etiologies?

↳ NOT ORAL (WONT DIGEST)

Manifestations: B12-deficiency anemia

- Classic symptoms of anemia
- Nervous system problems – Why?
 - Numbness, tingling and burning in feet and hands
 - Mental impairment, delusions, irritability
 - Depression

Aplastic Anemia

- RBC normal in size and shape but not enough of them because of failure of the red bone marrow
 - Bone marrow is fatty and unable to produce RBC



Aplastic Anemia

- Etiology

- congenital, industrial chemicals (benzenes, insecticides), infections, chemotherapy drugs, **idiopathic (in 2/3 cases)**

- Manifestations

- Onset is insidious
- fatigue, weakness, pallor, petechiae and ecchymosis, bleeding, increased risk of infection etc.

- Pathophysiology

- **exposure** → depression of bone marrow → decreased production of RBCs, WBCs, and platelets

→ from chemo or smthg else

- Manifestations

- Onset is insidious
- fatigue, weakness, pallor, petechiae and ecchymosis,

Aplastic anemia

Diagnosis	Treatment:	Prognosis:
• CBC – what results are expected? • Bone marrow biopsy	• correction of underlying cause • RBC transfusions, medications such as erythropoietin and corticosteroids • bone marrow transplant	• Poor if no treatment • 70% survive with <u>BMT</u> • Immuno-suppressive therapy- 50%

↳ bone marrow Transplant

CORTICOSTEROIDS

Hemolytic Anemia

- Excessive hemolysis or destruction of abnormal RBC
- Etiology: fragile or abnormal RBC are destroyed by the liver, spleen and bone marrow.
- Pathophysiology
 - Hemolysis of RBC produce excess amounts of byproducts – iron, bilirubin
 - Decreased circulating mature RBC, decreased oxygen carrying capacity
 - Increased clumping of RBC – embolisms/clots form

Manifestations

- Mild: classic anemia symptoms
- Severe: Jaundice, hemoglobinuria, *→ Blood in urine* hemoglobinemia, *→ Pink plasma less RBC on lab results* hypotension, dyspnea, splenomegaly, back and joint pain

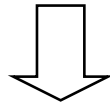
- Can you explain why these manifestations occur?

Hemolytic anemia Dx and Tx

- Additional diagnostic tests:
 - LDH (lactic dehydrogenase)— released when tissue is damaged.
- Coomb's test (antiglobulin test)
 - detects antibody on RBC that indicates immune system cause
- Treatment: determined by cause

Sickle Cell Disease

- Recessive inheritance
- Mutation in beta chains of hemoglobin (HbS)
sickled hemoglobin



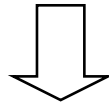
- When hemoglobin is deoxygenated, beta chains link together, forming long protein rods that make the cell “sickle”

Sickle Cell Disease- Recessive Inheritance

- *How many mutated genes need to be inherited to have:*
 - *Full sickle cell disease*
 - *Sickle cell carrier (Trait)*
- *If you carry the trait, will you have symptoms?*

Pathophysiology

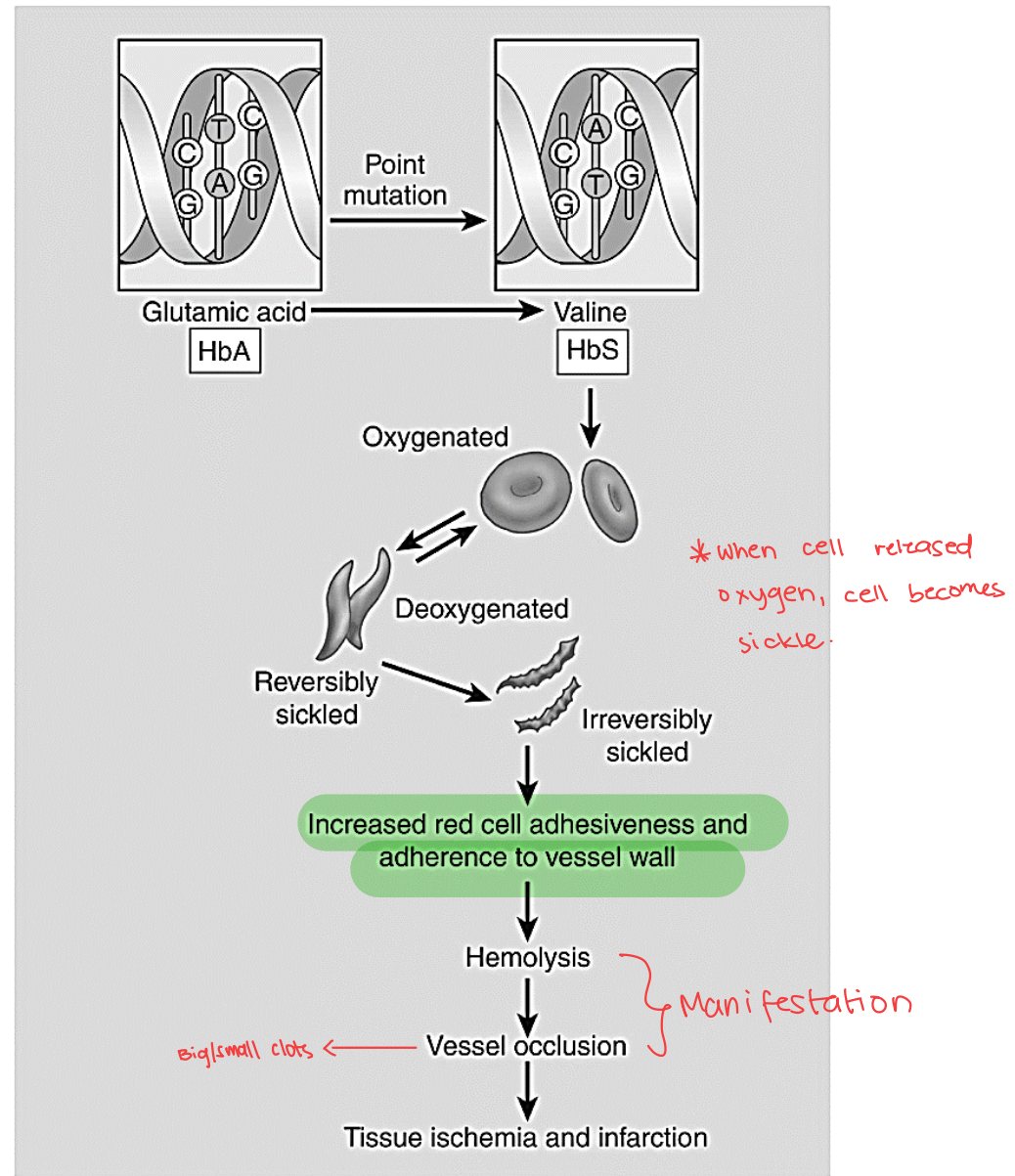
- Abnormal HbS
 - Mutation on β chain
 - Amino acid substitution



- When deoxygenated, HbS polymerizes

- HbS becomes irreversibly sickled

Figure 23-8 p574



Consequences of RBC sickling

Vessel occlusion

- Tissue ischemia → Acute pain
- Infarctions cause chronic damage to liver, spleen, heart, kidneys, eyes, bones
- Pulmonary infarction → acute chest syndrome
- Cerebral infarction → stroke

Chronic hemolytic anemia

- Sickled cells more likely to be destroyed
- Jaundice

Diagnosis

- Screening programs
 - Heel stick samples sent for electrophoresis to separate the HbS from HbF and HbA
- Neonatal Dx
 - Clinical findings
 - Hemoglobin electrophoresis
- Prenatal Dx
 - Amniocentesis

Sickle Cell Anemia Treatment

- No cure
- Avoid sickling triggers
- Treatment to reduce symptoms:
 - Pain control
 - Hydration etc.
 - Transfusion may be needed
- Antibiotics
- Immunization
- Hydroxyurea may be used to prevent complications and reduce pain crises

- **What are some triggers that are associated with sickling and vessel occlusion?**

- infection

- **Why blood transfusions?**
- **Why antibiotics and immunizations?**

Thalassemias

- Group of inherited disorders of hemoglobin synthesis
- Caused by absent or defective synthesis of α or β chains of adult hemoglobin and affects cell maturation
- Develop **hypochromic, microcytic** anemia
 - Decrease in Hgb and therefore RBC production
 - RBC are abnormal in shape and short lived.
 - Excessive hemolysis due to odd shape

Thalassemias

- Minor (trait carrier, mild anemia) or Major (full disease)
- Population affected:

Thalassemias

- Minor (trait carrier, mild anemia) or Major (full disease)
- Population affected:

Alpha	Beta	Alpha & Beta
• Asians	• Mediterranean population (Italy and Greece)	• Africans • African Americans

Thalassemias - pathogenesis

β - thalassemia

- Beta chains are defective and fewer produced
- Alpha chains unaffected and produced as usual.
- Leads to an imbalance (excess alpha, too little beta)
- Excess alpha chains form Heinz bodies which affects ability to mature, cell membranes are weak
- Leads to excessive hemolysis/destruction
- Ineffective hematopoiesis and hemolysis => $\uparrow$ erythropoietin => $\uparrow$ marrow mass

Thalassemias - pathogenesis

α - thalassemia

- Gene deletion => results in defective α chain synthesis
- 3-4 genes deleted causes unstable chains

Manifestations of β -thalassemia

- Usual signs of anemia as well as:
 - Bone deformities –why?
 - Splenomegaly and hepatomegaly- why?
 - Excessively high iron level- why?
 - Hgb levels can be very low, blood and tissue oxygen levels cause multiple problems
 - Increased thrombotic events (stroke etc)

Dx and Tx for β -thalassemia

- Diagnostic tests:
 - Microscopic exam of RBC – microcytic, hypochromic
 - Increased levels of erythropoietin
 - **Excessively high iron levels**
- Treatment
 - Bone marrow and stem cell transplantation, experimental gene manipulation are being explored
 - Splenectomy – how might this help?
 - Iron chelation therapy –purpose?
 - If severe, the client will **be transfusion dependent** starting from 6-9mths of age....Needed Q 3-4wks